

Bulk topology from truncated overcomplete Wannier modes

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We construct a half-filled reference state from truncated overcomplete Wannier modes and give a sufficient, volume-independent condition for its Chern number to remain one. We then distinguish this static statement from a possible deformation between imaginary-time filtering and monitored trajectories.

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I. THE UNTRUNCATED PROJECTOR

We consider the translation-invariant bulk, with two orbitals per unit cell, and the parent Hamiltonian of Ref. [1]. Its lower-band projector is

$$P_-(\mathbf{k}) = \frac{1}{2}[\mathbf{1} - \hat{\mathbf{n}}(\mathbf{k}) \cdot \boldsymbol{\sigma}], \quad \mathbf{n} = (\sin k_x, \sin k_y, \alpha - \cos k_x - \cos k_y), \quad 0 < \alpha < 2. \quad (1)$$

Here $\hat{\mathbf{n}} = \mathbf{n}/|\mathbf{n}|$ is a unit vector, not an operator. The half-filled ground state occupies P_- , with $\text{Ch}(P_-) = 1$ in our orientation convention. Choose $\tau_{A/B} = (1, \pm 1)^T/\sqrt{2}$ and write $|w_{\mathbf{r}\nu}\rangle = P_-|\mathbf{r}, \tau_\nu\rangle$. Completeness of the local trial states gives

$$\sum_{\mathbf{r}, \nu} |w_{\mathbf{r}\nu}\rangle \langle w_{\mathbf{r}\nu}| = P_-^2 = P_-. \quad \|w_{\mathbf{r}\nu}\|^2 = \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} \tau_\nu^\dagger P_-(\mathbf{k}) \tau_\nu = \frac{1}{2}. \quad (2)$$

The last equality follows from oddness of $n_x/|\mathbf{n}|$. Thus the normalized overcomplete Wannier (OW) columns $W_{\mathbf{r}\nu} = \sqrt{2} w_{\mathbf{r}\nu}$ satisfy $P_- = \frac{1}{2} \sum_{\mathbf{r}, \nu} W_{\mathbf{r}\nu} W_{\mathbf{r}\nu}^\dagger$. The factor 1/2 is essential: this is a tight frame, not an orthonormal basis. In the report's two-point ordering, $C_{ij} = \text{Tr}(\hat{\rho} \hat{c}_i^\dagger \hat{c}_j)$ and $\Gamma = C^T$; the ground state has $\Gamma = P_-$.

II. TRUNCATION AND QUANTIZATION

Let $g_\ell(\mathbf{r})$ be the indicator of $|x|, |y| \leq \ell$, where $\ell = n_{\text{shell}}$, and normalize $W_{\mathbf{r}\nu}^{(\ell)} = g_{\ell, \mathbf{r}} W_{\mathbf{r}\nu} / \|g_{\ell, \mathbf{r}} W_{\mathbf{r}\nu}\|$. Introduce the synthesis matrices $S_0 = (W_{\mathbf{r}\nu})/\sqrt{2}$, $S_\ell = (W_{\mathbf{r}\nu}^{(\ell)})/\sqrt{2}$. Since $S_0 S_0^\dagger = P_-$, $\|S_0\|_{\text{op}} = 1$. Setting $E_\ell = S_\ell - S_0$, $\epsilon_\ell = \|E_\ell\|_{\text{op}}$, we obtain

$$\begin{aligned} F_\ell &= S_\ell S_\ell^\dagger, & F_\ell - P_- &= S_0 E_\ell^\dagger + E_\ell S_0^\dagger + E_\ell E_\ell^\dagger, \\ \|F_\ell - P_-\|_{\text{op}} &\leq \delta_\ell, & \delta_\ell &= 2\epsilon_\ell + \epsilon_\ell^2. \end{aligned} \quad (3)$$

Small column errors alone would not give a volume-independent bound. Here translation symmetry does: write $W_\nu = W_{\mathbf{0}\nu}$, $M = \max_\nu \|W_\nu\|_1$ and $a_p = \max_\nu \|(1 - g_\ell)W_\nu\|_p$, where the norms include both orbitals. For $a_2 < 1$, normalization and the Fourier triangle inequality imply

$$\epsilon_\ell = \sup_{\mathbf{k}} \|E_\ell(\mathbf{k})\|_{\text{op}} \leq b_\ell := a_1 + \left[(1 - a_2^2)^{-1/2} - 1 \right] M \longrightarrow 0. \quad (4)$$

Indeed, each normalized-column difference has ℓ^1 norm at most b_ℓ ; the two Bloch columns, each divided by $\sqrt{2}$, have combined Frobenius norm at most b_ℓ . Exponential OW localization makes both tails vanish as $\ell \rightarrow \infty$, while M is finite.

Assume $\delta_\ell < 1/2$. Along $F(u) = P_- + u(F_\ell - P_-)$, spectral stability gives

$$\text{spec } F(u) \subset [-u\delta_\ell, u\delta_\ell] \cup [1 - u\delta_\ell, 1 + u\delta_\ell], \quad \Pi(u) = \mathbf{1}_{(1/2, \infty)}(F(u)), \quad 0 \leq u \leq 1. \quad (5)$$

The occupation gap stays open. Equivalently, $\Pi(u, \mathbf{k}) = (2\pi i)^{-1} \oint_{|z-1|=1/2} (z\mathbf{1} - F(u, \mathbf{k}))^{-1} dz$, with positive contour orientation. This defines a smooth, periodic rank-one projector. Its invariant is

$$\text{Ch}(\Pi(u)) = \frac{1}{2\pi i} \int_{\text{BZ}} d^2k \, \text{tr}(\Pi(u)[\partial_{k_x}\Pi(u), \partial_{k_y}\Pi(u)]). \quad (6)$$

On overlapping momentum patches, $|v_a\rangle = e^{i\theta_{ab}}|v_b\rangle$ and $\langle v_a|dv_a\rangle = \langle v_b|dv_b\rangle + i d\theta_{ab}$. Stokes' theorem identifies Eq. (6) with the integer winding $(2\pi)^{-1} \oint d\theta_{ab}$, summed over oriented patch boundaries. It is also continuous in u , hence constant:

$$\boxed{\epsilon_\ell < \sqrt{3/2} - 1 \implies \text{Ch}(\Pi_\ell) = \text{Ch}(P_-) = 1, \quad \Pi_\ell = \Pi(1).} \quad (7)$$

Uniform truncation preserves momentum diagonality. The real-space projector is exponentially local because it is the gapped spectral projection of a finite-range operator. Its infinite-bulk three-sector invariant is the same integer [2]; a finite disk still has finite-radius corrections. This proves the claim for sufficiently large finite ℓ , not specifically for $\ell = 1$. Nor is F_ℓ itself a projector or an occupation matrix: its eigenvalues can exceed one. The reference Hamiltonian $h_\ell = \mathbf{1} - 2F_\ell$ has half-filled ground-state projector Π_ℓ .

III. IMAGINARY TIME AND THE MONITORED CIRCUIT

For a finite system, normalized $e^{-\beta \hat{H}_\ell} |\psi_0\rangle\rangle$ approaches this ground state as $\beta \rightarrow \infty$, provided the initial half-filled state has nonzero ground-state overlap; here $\hat{H}_\ell = \sum_{ij} (h_\ell)_{ij} \hat{c}_i^\dagger \hat{c}_j$. There is a useful connection to occupation measurements. With $\hat{N}_{j,\pm} = \hat{\chi}_{j,\pm}^\dagger \hat{\chi}_{j,\pm}$ for normalized truncated modes, the penalties $\hat{q}_{j,-} = \hat{\mathbf{1}} - \hat{N}_{j,-}$ and $\hat{q}_{j,+} = \hat{N}_{j,+}$ are projectors, so

$$e^{-\lambda \hat{q}} = \hat{\mathbf{1}} + (e^{-\lambda} - 1)\hat{q}, \quad e^{-\lambda \hat{q}_{j,-}} \xrightarrow{\lambda \rightarrow \infty} \hat{N}_{j,-}, \quad e^{-\lambda \hat{q}_{j,+}} \xrightarrow{\lambda \rightarrow \infty} \hat{\mathbf{1}} - \hat{N}_{j,+}. \quad (8)$$

These are precisely the desired-outcome, no-correction branches. Trotterizing the positive penalty $\hat{H}_{\text{pen}} = \sum_{j,\sigma} \hat{q}_{j,\sigma}$ gives $e^{-\beta \hat{H}_{\text{pen}}} = \lim_{M \rightarrow \infty} (\prod_{j,\sigma} e^{-\beta \hat{q}_{j,\sigma}/M})^M$ at fixed finite volume and β . Its one-body matrix is $2(F_+^{(\ell)} - F_-^{(\ell)})$, not generally h_ℓ . Here $F_-^{(\ell)} = F_\ell$ and $F_+^{(\ell)}$ is the analogous upper-band frame operator. If $\|F_\pm^{(\ell)} - P_\pm\|_{\text{op}} \leq \delta_\pm$ and $\delta_+ + \delta_- < 1$, its ground-state projector is likewise connected to P_- without closing the gap. This is a static spectral argument, distinct from dissipative preparation [3].

We may therefore conjecture that typical late-time trajectories remain in this deformed topological sector. However, weak-filter Trotterization does not justify the strong-filter limit at fixed step size: the penalties do not commute. An arbitrary record also contains the correction branches $\hat{\chi}_{j,-}^\dagger$ and $\hat{\chi}_{j,+}$, which change physical charge; projective branches are singular. Imaginary time under $\hat{H}_\ell$ preserves charge. A deformation to an arbitrary record is consequently not automatic. It would require control of bulk locality and a spectral or mobility gap along a record-dependent interpolation, with any charge-changing localized defects treated explicitly. Equation (7) proves topology of the reference state; extending it to the conditional ensemble is the open step.

[1] A. Bhuiyan, H. Pan, and C.-M. Jian, *Phys. Rev. Research* **8**, 023147 (2026).

[2] A. Kitaev, *Ann. Phys.* **321**, 2 (2006), Appendix C.3.

[3] M. Goldstein, *SciPost Phys.* **7**, 067 (2019).